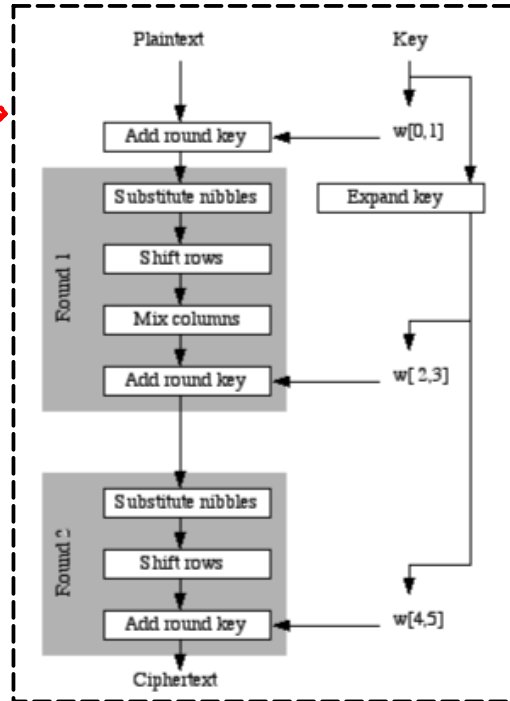
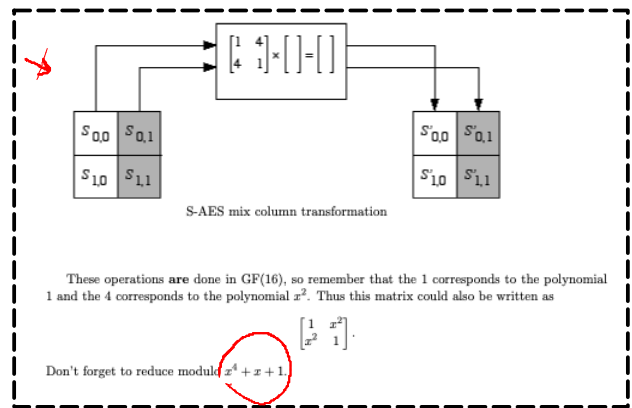
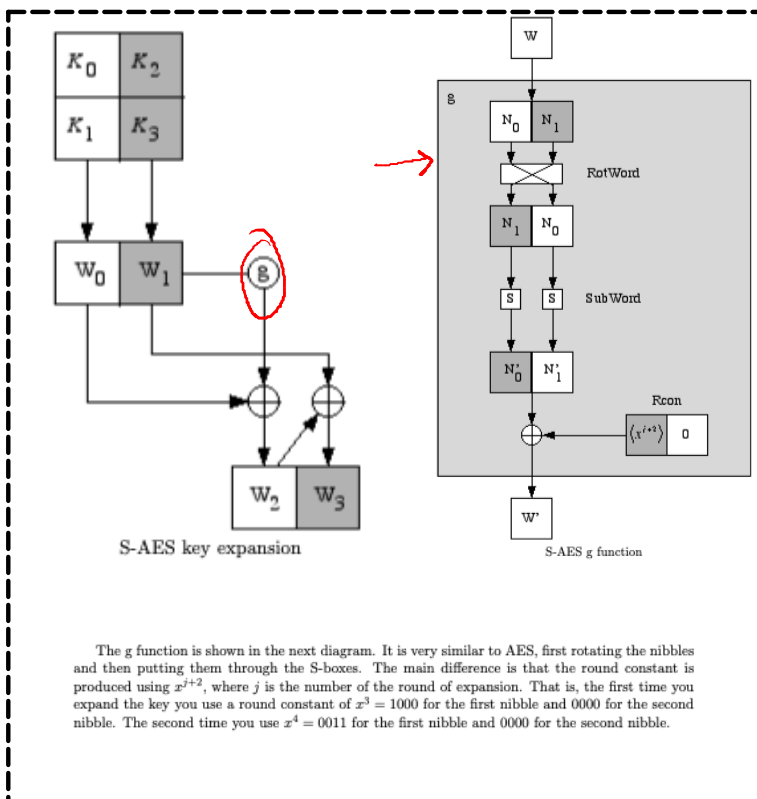
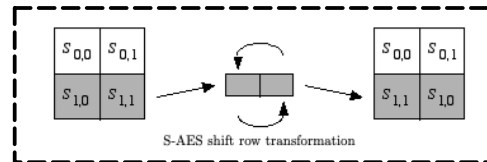


nibble	S-box(nibble)	nibble	S-box(nibble)
0000	1001	1000	0110
0001	0100	1001	0010
0010	1010	1010	0000
0011	1011	1011	0011
0100	1101	1100	1100
0101	0001	1101	1110
0110	1000	1110	1111
0111	0101	1111	0111



Exercise: Use the key 1010 0111 0011 1011 to encrypt the plaintext "ok" as expressed in ASCII, that is 0110 1111 0110 1011. The designers of S-AES got the ciphertext 0000 0111 0011 1000. Do You?

Key Expansion

1010	0011
0111	1011

w_0 w_1

$$w_2 = g(w_1) \oplus w_0$$

$$w_3 = w_2 \oplus w_1$$

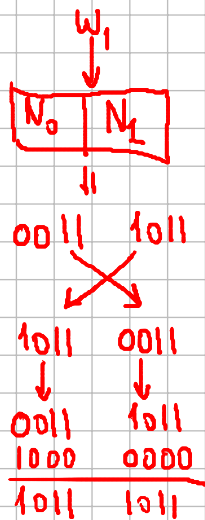
$$w_4 = g(w_3) \oplus w_2$$

$$w_5 = w_4 \oplus w_3$$

$$w_0 = 1010\ 0111$$

$$w_1 = 0011\ 1011$$

$$g(w_1) = ?$$

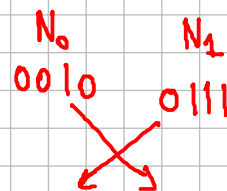
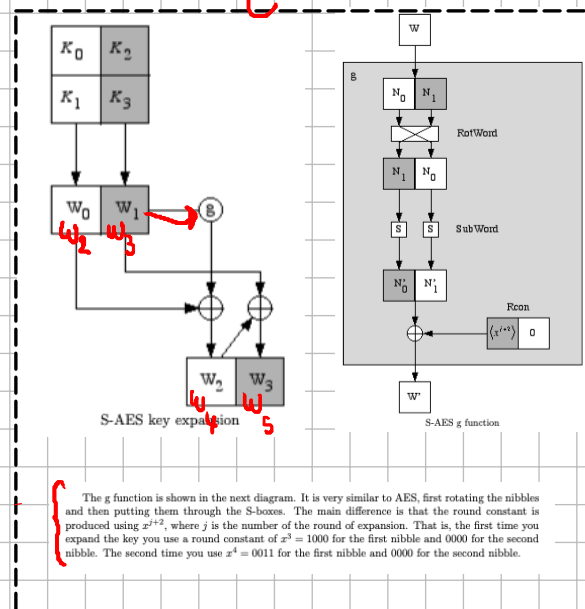


$$w_2: \begin{array}{r} 1011\ 1011 \\ 1010\ 0111 \\ \hline 0001\ 1100 \leftarrow w_2 \end{array}$$

$$w_3 = w_2 \oplus w_1 \rightarrow \begin{array}{r} 0001\ 1100 \\ 0011\ 1011 \\ \hline 0010\ 0111 \leftarrow w_3 \end{array}$$

$$w_4: \begin{array}{r} 0110\ 1010 \\ 0001\ 1100 \\ \hline 0111\ 0110 \leftarrow w_4 \end{array}$$

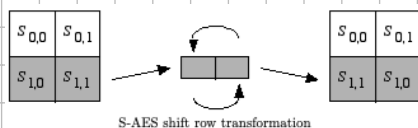
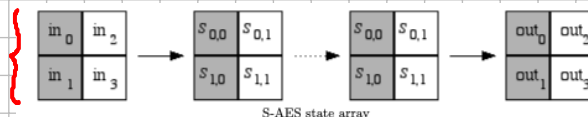
$$w_5: \begin{array}{r} 0111\ 0110 \\ 0110\ 0111 \\ \hline 0101\ 0001 \end{array}$$



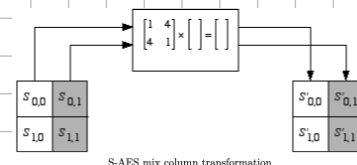
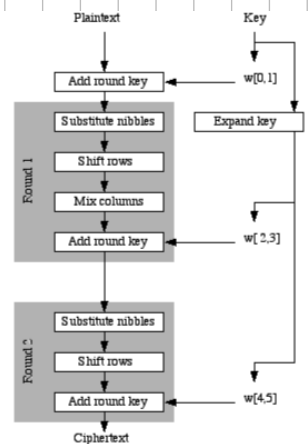
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0001	0100	1001	0010
0010	1010	1010	0000
0011	1011	1011	0011
0100	1101	1100	1100
0101	0001	1101	1110
0110	1000	1110	1111
0111	0101	1111	0111

$$w_4: \begin{array}{r} 0111\ 0110 \\ 0110\ 0111 \\ \hline 0101\ 0001 \leftarrow w_4 \end{array}$$

W0 = 1010 0111
W1 = 0011 1011
W2 = 0001 1100
W3 = 0010 0111
W4 = 0111 0110
W5 = 0101 0001



nibble	S-box(nibble)	nibble	S-box(nibble)
0000	1001	1000	0110
0001	0100	1001	0010
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0011	1011	1011	0011
0100	1101	1100	1100
0101	0001	1101	1110
0110	1000	1110	1111
0111	0101	1111	0111



These operations are done in GF(16), so remember that the 1 corresponds to the polynomial 1 and the 4 corresponds to the polynomial x^2 . Thus this matrix could also be written as

$$\begin{bmatrix} 1 & x^2 \\ x^2 & 1 \end{bmatrix}$$

Don't forget to reduce modulo $x^4 + x + 1$.

PT = 0110 1111 0110 1011

$$\begin{bmatrix} 0110 & 0110 \\ 1111 & 1011 \end{bmatrix} \oplus \begin{bmatrix} 1010 & 0011 \\ 0111 & 1011 \end{bmatrix} = \begin{array}{c|c} 1100 & 0101 \\ \hline 1000 & 0000 \end{array} \rightarrow S$$

$$\begin{array}{c|c} 1100 & 0001 \\ \hline 1001 & 0110 \end{array} \xleftarrow{SR} \begin{array}{c|c} 1100 & 0001 \\ \hline 0110 & 1001 \end{array}$$

$$\begin{bmatrix} 1 & x^2 \\ x^2 & 1 \end{bmatrix} \begin{bmatrix} x^3 + x^2 & 1 \\ x^3 + 1 & x^2 + x \end{bmatrix} = \begin{bmatrix} x^3 + x^2 + x^5 + x^3 & 1 + x^4 + x^3 \\ x^5 + x^4 + x^3 + 1 & x^2 + x^2 + x \end{bmatrix}$$

$$= \begin{bmatrix} x^5 + x^3 & x^4 + x^3 + 1 \\ x^5 + x^4 + x^3 + 1 & x \end{bmatrix}$$

$$= \begin{bmatrix} x^3 + x^2 + x & x^3 + x \\ x^3 + x^2 & x \end{bmatrix}$$

$$= \begin{array}{c|c} 1110 & 1010 \\ \hline 1100 & 0010 \end{array}$$

$$\oplus \begin{bmatrix} 0001 & 0010 \\ 1100 & 0111 \end{bmatrix} = \begin{bmatrix} 1111 & 1000 \\ 0000 & 0101 \end{bmatrix}$$

$$\begin{array}{c|c} x^5 + x^3 & x^4 + x^3 + 1 \\ \hline x^5 + x^4 + x^3 + 1 & x \end{array}$$

$$\begin{array}{c|c} x^5 + x^3 + 1 & x^4 + x^3 + 1 \\ \hline x^5 + x^4 + x^3 + 1 & 1 \end{array}$$

$$\begin{array}{c|c} x^5 + x^4 + x^3 + 1 & x^4 + x^3 + 1 \\ \hline x^5 + x^4 + x^3 + 1 & x^3 + x^2 \end{array}$$

1111	1000
0000	0101



0111	0110
1001	0001

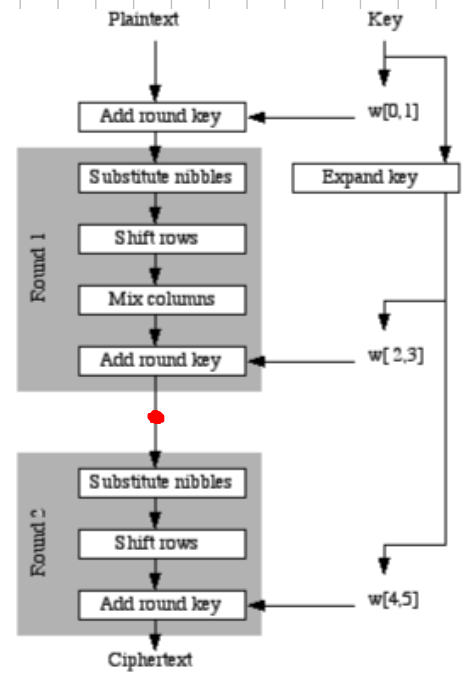


0111	0110
0001	1001

⊕

nibble	S-box(nibble)	nibble	S-box(nibble)
0000	1001	1000	0110
0001	0100	1001	0010
0010	1010	1010	0000
0011	1011	1011	0011
0100	1101	1100	1100
0101	0001	1101	1110
0110	1000	1110	1111
0111	0101	1111	0111

W0 = 1010 0111
W1 = 0011 1011
W2 = 0001 1100
W3 = 0010 0111
W4 = 0111 0110
W5 = 0101 0001



0000 0111 0011 1000 ✓

0111	0101
0110	0001

=

0000	0011
0111	1000